HK IMO Training 2019-2020
Problem Set 1 Suggested Solutions

Problem 1. Let n be a positive integer such that 24 divides $n+1$. Prove that the sum of the positive divisors of n is divisible by 24.

Solution. If d is some divisor of n, then $\frac{n}{d}$ is also a divisor of n. Since $n \equiv 2 \pmod 3$, it is not a perfect square. Then we can write

$$\sigma(n) = \sum_{\substack{d|n \\ d^2<n}} \left(d + \frac{n}{d}\right) = \sum_{\substack{d|n \\ d^2<n}} \frac{d^2+n}{d},$$

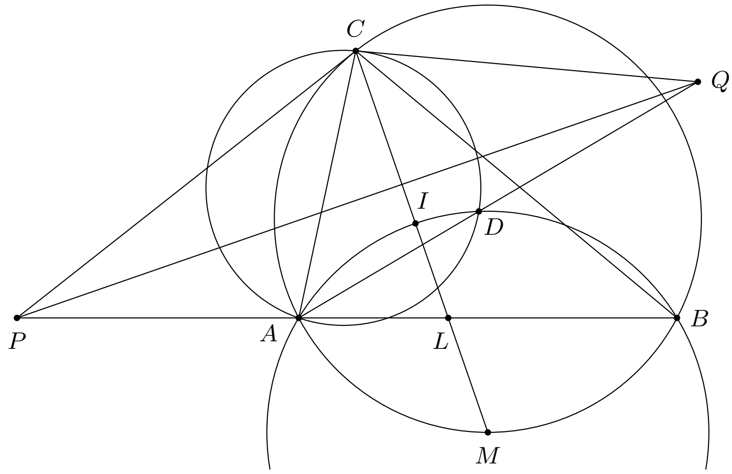
where $\sigma(n)$ denotes the sum of the positive divisors of n. Notice that $n \equiv 2 \pmod 3$ and $d^2 \equiv 1 \pmod 3$. So $3 \mid \sigma(n)$. On the other hand, $n \equiv 7 \pmod 8$ and $d^2 \equiv 1 \pmod 8$. So $8 \mid \sigma(n)$. It follows that $24 \mid \sigma(n)$. $\square$

Problem 2. Let $P(x)$ be a polynomial with integer coefficients. Prove that if there exist integers a, b with $|P(a)| = |P(b)| = 1$ and $|a-b| \geqslant 3$, then $P(x)$ has no integer roots.

Solution. Suppose on the contrary that $P(x)$ has an integer root c. Since $|a-b| \geqslant 3$, without loss of generality, we may assume $|a-c| > 1$. As $P(x)$ is a polynomial with integer coefficients, we have $a - c \mid P(a) - P(c)$. But $|P(a) - P(c)| = |P(a)| = 1$. This contradicts the assumption that $|a-c| > 1$. So $P(x)$ has no integer roots. $\square$

Problem 3. Point D is taken inside $\triangle ABC$ with $AC \neq BC$ such that $\angle ADB = 90^\circ + \frac{1}{2}\angle ACB$. The tangents at point C to the circumcircles of $\triangle ABC$ and $\triangle ADC$ meet the lines AB and AD respectively at points P and Q. Prove that line PQ bisects $\angle BPC$.

Solution.



Denote by I the incentre of $\triangle ABC$. Let the line CI meet the side AB at point L and the circumcircle of $\triangle ABC$ again at point M. Point M is the centre of the circumcircle ω of $\triangle AIB$. Since $\angle ADB = 90^\circ + \dfrac{1}{2}\angle ACB = \angle AIB$, point D also lies on ω.

Since $PC^2 = PA \cdot PB$ and $QC^2 = QA \cdot QD$, points P and Q have equal powers with respect to the degenerate circle C and ω, so PQ is the radical axis of these two circles. Hence PQ is perpendicular to the line connecting their centres, so $PQ \perp CL$. Since

$$\measuredangle PCL = \measuredangle PCA + \measuredangle ACL = \measuredangle CBA + \measuredangle LCB = \measuredangle CLP,$$

we get $PL = PC$. It follows that PQ bisects $\angle CPL$. $\square$

Problem 4. In a certain city there are $n \geqslant 2$ married couples numbered from 1 to n. On a certain day, each man congratulated two women from couples whose numbers differ by one. It turned out that for any two men from couples whose numbers differ by one, there is a woman who received congratulations from both of them. Prove that at least one man congratulated his wife.

Solution. Assume the contrary: none of the men congratulated his wife. Call a man from kth couple *leftside*, if he congratulated women from the couples with the numbers greater than k, otherwise call him *rightside*. Clearly the man from the first couple is leftside and the man from the nth couple is rightside. Hence there is a number i, $1 \leqslant i \leqslant n-1$, such that the man from ith couple is leftside while the man from the $(i+1)$st couple is rightside. The numbers of couple of these men differ by one, but they congratulated four different women — a contradiction to the problem condition. Consequently at least one man congratulated his wife. $\square$